

ChatGPT’s release, with a one standard deviation increase in LLM exposure reducing the probability of posting any vacancy by 1.0 percentage point. This reduction emerges gradually and becomes significant after approximately 10 months, mirroring the timeline of employment stock adjustments. When we examine the composition of roles driving this reduction, we find that technical positions and creative roles experience the sharpest declines. Sales representative positions, by contrast, show a statistically significant increase, suggesting that customer-facing roles requiring direct interpersonal interaction remain robust to LLM-based automation.

In addition to within-firm dynamics, our occupation-level analysis reveals how labor demand shifts for entire job categories regardless of firm exposure. A one standard deviation increase in LLM exposure is associated with a 3.9 percent reduction in posting volumes, accompanied by £495 (1.1 percent) salary declines. The apparent inconsistency of declining posted salaries for exposed occupations alongside increasing firm-level compensation shows how aggregation could mask opposing forces. While LLM exposure drives down market wages for specific occupations, within-firm rebalancing toward more senior positions pushes average firm-level wages in the opposite direction.

Our heterogeneity analysis reveals that the observed labor market disruptions are almost entirely driven by high-remuneration segments of the labor market. This concentration of effects in higher-wage segments contradicts historical automation patterns that predominantly displaced routine middle-skill work, and suggests that LLMs might lead to a compression of the wage distribution.

Our findings carry important implications for understanding technological displacement and designing policy responses. The concentration of job losses among entry-level positions disrupts traditional skill development pathways where workers master progressively complex tasks through hands-on experience. Without junior roles serving as training grounds, firms may struggle to develop senior talent internally while new entrants face shrinking opportunities to gain experience (Ide, 2025). The freeze in hiring followed by employment stock changes suggests that forward-looking indicators like job postings may provide early warning signals for policymakers. The sharp reductions in technical and analytical roles alongside resilience in sales positions indicates that human interaction and relationship-building skills remain valuable complements to LLM capabilities, suggesting that retraining programs should emphasize these interpersonal competencies. Policy responses